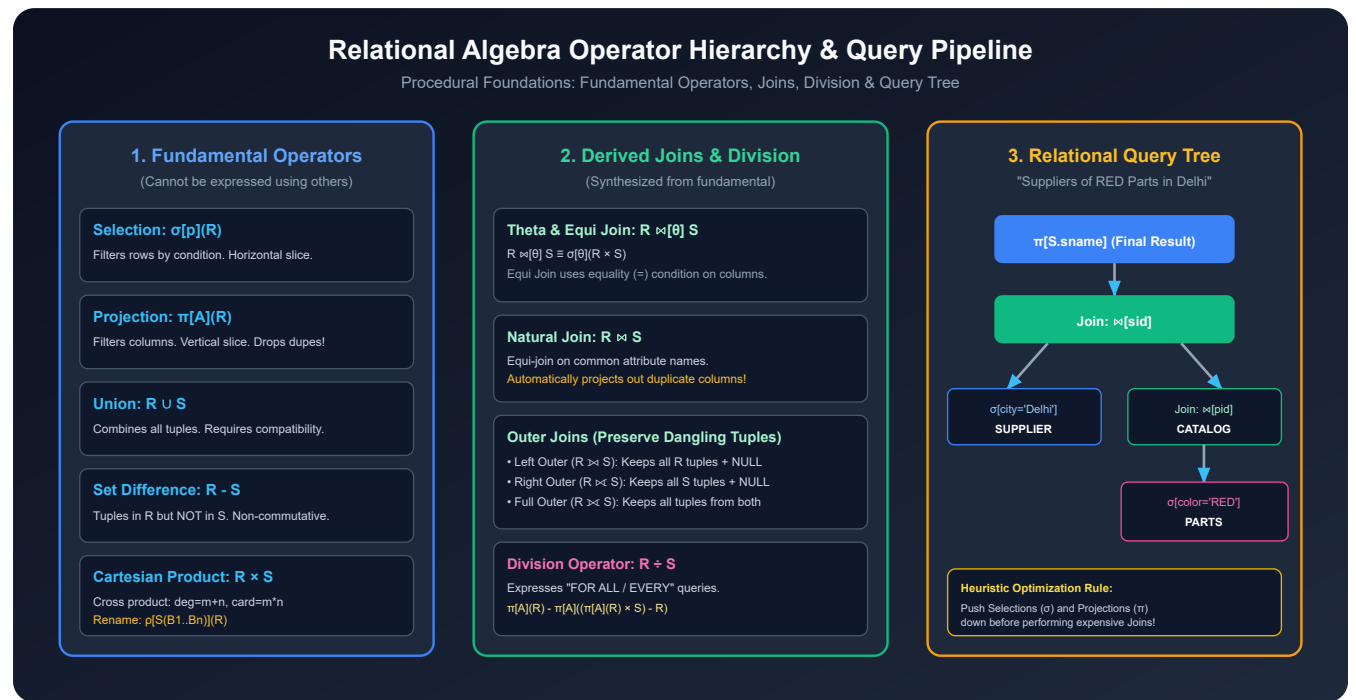


# Module 01: Relational Algebra Complete Operators and Solved Numericals



Procedural Query Foundation: Relational Algebra specifies WHAT data to retrieve and the exact procedural STEP-BY-STEP pipeline.

## 1. Fundamental Operators

Fundamental operators are atomic. The 5 core operators are Selection ( $\sigma$ ), Projection ( $\pi$ ), Union ( $\cup$ ), Set Difference ( $-$ ), and Cartesian Product ( $\times$ ).

| Operator          | Symbol        | Dimension         | Key Behavior                                                                                 | Closure Property                       |
|-------------------|---------------|-------------------|----------------------------------------------------------------------------------------------|----------------------------------------|
| Selection         | $\sigma_p(R)$ | Horizontal (Rows) | Filters tuples based on predicate condition p.                                               | Preserves degree, reduces cardinality. |
| Projection        | $\pi_A(R)$    | Vertical (Cols)   | Extracts specified columns and automatically eliminates duplicate rows.                      | Reduces degree to $ A $ .              |
| Union             | $R \cup S$    | Row Concatenation | Combines tuples. Requires Union Compatibility (same degree & domains).                       | Removes duplicates.                    |
| Set Difference    | $R - S$       | Tuple Subtraction | Tuples present in R but absent in S. Non-commutative ( $R - S \neq S - R$ ).                 | Union compatible.                      |
| Cartesian Product | $R \times S$  | All Combinations  | Pairs every tuple of R with every tuple of S.<br>$\text{deg} = m + n, \text{card} = m * n$ . | No condition needed.                   |

## 2. Derived Joins & Division

Natural Join ( $R \bowtie S$ ) performs an equality join on common columns and drops redundant duplicate columns. Outer joins (Left, Right, Full) preserve non-matching dangling tuples by padding missing values with NULLs.

Division Operator ( $R \div S$ ): Formulated as  $\pi_A(R) - \pi_A((\pi_A(R) \times S) - R)$ . Used to answer "FOR ALL / EVERY" queries.

### 3. Solved AKTU Numerical Schema

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Suppliers-Parts Database: SUPPLIERS(sid, sname, address), PARTS(pid, pname, color), CATALOG(sid, pid, cost).

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-- Find names of suppliers who supply some RED part:  
 $\pi[\text{sname}](\text{SUPPLIERS} \bowtie \pi[\text{sid}](\text{CATALOG} \bowtie \sigma[\text{color} = \text{'RED'}](\text{PARTS})))$   
  
-- Find suppliers who supply EVERY part:  
 $\pi[\text{sid}, \text{pid}](\text{CATALOG}) \div \pi[\text{pid}](\text{PARTS})$ 
```